

ImFormer supplement - Cross Attention

Seq-to-seq learning:

For target tokens $t_1, \dots, t_m$ learn implicit model

(after Nesterov Rademacher descent to reduce

average Renyi α -entropy of query updates across keys)

$$s_i = \psi \left(\overset{k_i}{\underset{\parallel}{K}} s_i + \overset{q_i}{\underset{\parallel}{Q}} t_i \right)$$

$$\varepsilon = \underset{\underset{\parallel}{V}}{V} s_i + \underset{\underset{\parallel}{q'}}{Q'} t_i$$

Similarly for source tokens $\tilde{t}_1, \dots, \tilde{t}_n$ learn

$$\tilde{s}_i = \psi \left(\tilde{\underset{\parallel}{K}} \tilde{s}_i + \tilde{\underset{\parallel}{Q}} \tilde{t}_i \right)$$

$$\tilde{\varepsilon} = \tilde{\underset{\parallel}{V}} \tilde{s}_i + \tilde{\underset{\parallel}{Q'}} \tilde{t}_i$$

$$\alpha_{ij} \triangleq \text{softmax}_j \text{ score}(q'_i, k_j), \quad j_i^* = \arg \max_j \alpha_{ij}$$

$$\text{similarly } \tilde{\alpha}_{ij} \propto \text{score}(\tilde{q}'_i, \tilde{k}_j)$$

$$h_i \triangleq \left[\sum_{j=1}^m \alpha_{ij} v_j \right]_{\Theta}, \text{ similarly } \tilde{h}_j \triangleq \left[\sum_{k=1}^n \tilde{\alpha}_{jk} v_k \right]_{\tilde{\Theta}}$$

where $[\]_{\Theta}$ denotes a FeedForward network applied elementwise.

$$\beta_{ij} \triangleq \text{softmax score}(h_i, k_j), \quad j_i^\beta = \arg \max_j \beta_{ij}$$

$$\text{similarly } \tilde{\beta}_{ij} \propto \text{score}(\tilde{h}_i, \tilde{k}_j)$$

$$\text{output } o_{i+1} \triangleq t_{j_i^\beta}$$

$$\text{similarly } \tilde{o}_{i+1} \triangleq \tilde{t}_{j_i^{\tilde{\beta}}}$$

Learn Θ to minimize perplexity $(o_2, \dots, o_{m+1})$

similarly learn $\tilde{\Theta}$ to minimize perplexity $(\tilde{o}_2, \dots, \tilde{o}_{n+1})$

Assume the h_i ($\tilde{h}_j$) have been updated multiple times.

$$\text{Define } \gamma_{ij} = \text{softmax score}_{j \in [n]}(h_i, \tilde{h}_j), \quad i \in [m]$$

$$\text{and } j_i^\gamma = \arg \max_{j \in [n]} \gamma_{ij}$$

Define the translation of t_i ,

$$\text{to be } \tilde{t}_{j_i^\gamma} \quad \text{where } \tilde{l} \triangleq j_i^\gamma$$

Cross attention between updates h_i and $\tilde{h}_j$.